PROJECT REPORT

ON

DESIGN AND ANALYSIS OF 1-BIT FULL ADDER USING CADENCE VIRTUOSO

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ABSTRACT

This project presents the design and simulation of a 1-bit Full Adder circuit using Cadence Virtuoso EDA (Electronic Design Automation) tool. Five different circuit topologies for the 1-bit Full Adder are designed, simulated, and compared with respect to their power consumption, propagation delay, and transistor count. The circuits are implemented at the transistor level using CMOS technology nodes within the Cadence Virtuoso Schematic Editor, and transient analysis is performed using the Spectre simulator.

The five design approaches evaluated are: (1) Standard CMOS static logic, (2) Hybrid CMOS full adder, (3) Pass transistor logic full adder, (4) Gate Diffusion Input (GDI)-based full adder, and (5) Hybrid GDI full adder. Delay calculations are performed for both the Sum and Carry-out (Cout) outputs with respect to varying Cin input transitions. The results demonstrate that hybrid and GDI-based topologies offer significant improvements in delay and power over the conventional static CMOS approach, making them attractive for low-power, high-speed VLSI applications.

ACKNOWLEDGEMENTS

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I would also like to express my gratitude to the Department of SENSE, Vellore Institute of Technology, Chennai, for providing all the necessary resources and a conducive environment for learning, research, and experimentation. My heartfelt thanks also go to the lab assistants and technical staff for their timely support and assistance during the practical work.

Lastly, I would like to thank my family and friends for their unwavering encouragement, motivation, and moral support throughout this journey. This project would not have been possible without the collective support of all the individuals mentioned above.

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INTRODUCTION

A full adder is one of the fundamental building blocks of digital arithmetic circuits. It performs the addition of three single-bit binary numbers — two operand bits (A and B) and a carry-in bit (Cin) — producing a Sum and a Carry-out (Cout). Full adders are widely used in arithmetic logic units (ALUs), digital signal processors, and microprocessors as the basic cell of multi-bit binary adders.

With the continued scaling of CMOS technology, reducing power dissipation and propagation delay while maintaining circuit robustness has become a primary challenge in modern VLSI design. The design of an efficient full adder at the transistor level is therefore of critical importance. Different circuit styles — from conventional static CMOS to pass transistor logic and GDI (Gate Diffusion Input) techniques — offer different trade-offs in terms of speed, power, and area.

This project explores five distinct transistor-level implementations of a 1-bit full adder using Cadence Virtuoso, one of the industry-leading EDA platforms for custom IC design. Each circuit is designed schematically, simulated with transient analysis, and analyzed for power dissipation and propagation delay. The objective is to identify the optimal design in terms of performance metrics for low-power VLSI applications.

The key contributions of this project are:

1. Transistor-level schematic design of five full adder topologies in Cadence Virtuoso.
2. Transient simulation and power analysis using Spectre simulator.
3. Propagation delay characterization for Sum and Cout outputs with varying Cin input transitions.
4. Comparative evaluation of all five circuits highlighting the best-performing topology.

BACKGROUND AND THEORY

2.1 Full Adder — Operation and Truth Table

A 1-bit full adder accepts three 1-bit inputs: A, B, and Cin (carry-in), and produces two 1-bit outputs: Sum and Cout (carry-out). The Boolean equations governing the full adder outputs are:

$$\mathbf{Sum = A \oplus B \oplus Cin}$$

$$\mathbf{Cout = (A \cdot B) + (B \cdot Cin) + (A \cdot Cin)}$$

The truth table for a 1-bit full adder is given below:

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Table 1: Truth Table of a 1-Bit Full Adder

2.2 CMOS Technology Overview

- Complementary Metal-Oxide-Semiconductor (CMOS) technology is the dominant technology used in modern integrated circuits. CMOS uses both PMOS (p-type) and NMOS (n-type) transistors in complementary pairs to implement logic functions. The key advantages of CMOS are low static power dissipation, high noise margins, and scalability.
- In a standard CMOS gate, the pull-up network (PUN) composed of PMOS transistors drives the output to VDD, while the pull-down network (PDN) composed of NMOS transistors drives the output to GND. The duality between these networks ensures that the output is always driven to a valid logic level, eliminating static current paths in steady state.

2.3 Design Topologies Overview

- Five full adder topologies are explored in this project:
- Static CMOS Full Adder: Uses standard complementary CMOS gates with PMOS pull-up and NMOS pull-down networks. High transistor count but reliable operation.
- Hybrid CMOS Full Adder: Combines standard CMOS gates with pass transistor logic to reduce transistor count and improve speed.
- Pass Transistor Logic (PTL) Full Adder: Uses pass transistors (NMOS or CMOS transmission gates) to pass logic values directly, reducing transistor count.
- GDI-Based Full Adder: Uses the Gate Diffusion Input (GDI) technique, a method for implementing complex logic functions with very few transistors.
- Hybrid GDI Full Adder: A combination of GDI cells and standard CMOS logic to achieve the lowest power and delay.

DESIGN METHODOLOGY

Tool Used: Cadence Virtuoso

Cadence Virtuoso is a comprehensive EDA (Electronic Design Automation) platform widely used in the semiconductor industry for custom analog, mixed-signal, and digital integrated circuit design. The key tools within the Virtuoso environment utilized in this project are:

- Virtuoso Schematic Editor: Used to draw the transistor-level schematics of all five full adder circuits.
- Analog Design Environment (ADE): Used to configure and run simulations.
- Spectre Simulator: The primary SPICE-compatible simulator used for transient analysis and power calculations.
- Virtuoso Waveform Viewer: Used to visualize transient waveforms and measure delay.

Simulation Setup

All circuits are designed using a standard CMOS technology PDK (Process Design Kit). The simulation parameters are set as follows:

- Supply voltage (VDD): 1.8 V or 3.3 V (technology-dependent)
- Analysis type: Transient analysis
- Input waveforms: Piecewise-linear voltage sources with rising and falling edges applied to A, B, and Cin
- Power calculation: Measured via Spectre's transient power analysis at the VDD supply
- Delay measurement: Propagation delay measured between input transitions and corresponding output transitions at 50% voltage crossings

For delay measurements, two output paths are analyzed separately: Cin to Sum (Cin vs Sum) and Cin to Cout (Cin vs Cout). Both rising and falling edge delays are recorded for completeness.

CIRCUIT 1: STATIC CMOS FULL ADDER

The first circuit is a conventional static CMOS implementation of the 1-bit full adder. In this design, all logic is implemented using complementary PMOS (pull-up) and NMOS (pull-down) transistors connected in standard CMOS gate configurations. This topology provides full rail-to-rail swing and high noise immunity.

Schematic

The schematic of Circuit 1 is implemented in Cadence Virtuoso with inputs A, B, and Cin, and outputs Sum and Cout. PMOS transistors (labeled P1–P12) form the pull-up network, while NMOS transistors (labeled N1–N12) form the pull-down network. The circuit uses a total of 24 transistors to implement the full adder function using two CMOS XOR gates and a carry generation network.

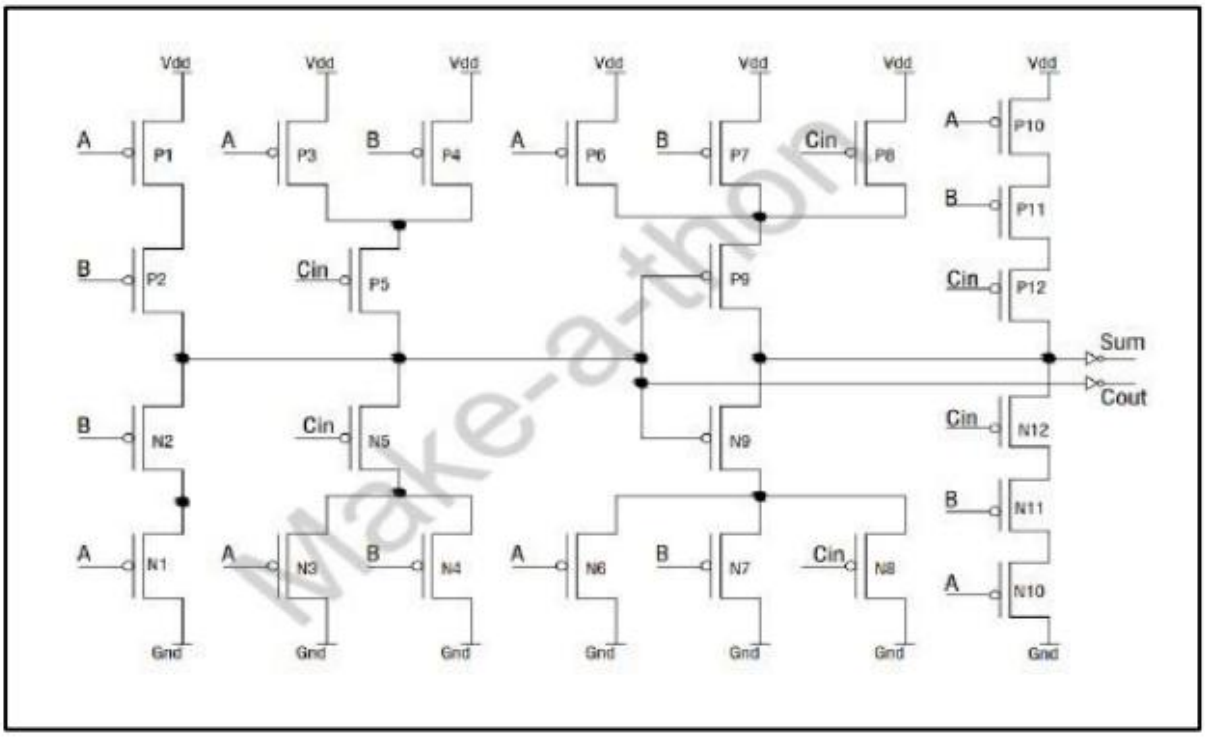

Figure 1: Cadence Virtuoso Schematic — Circuit 1 (Static CMOS Full Adder)

Simulation

The transient simulation of Circuit 1 is performed using the Spectre simulator. The input waveforms for A, B, and Cin are applied as time-varying voltage sources. The simulation captures the dynamic switching behavior of the Sum and Cout outputs in response to all possible input combinations.

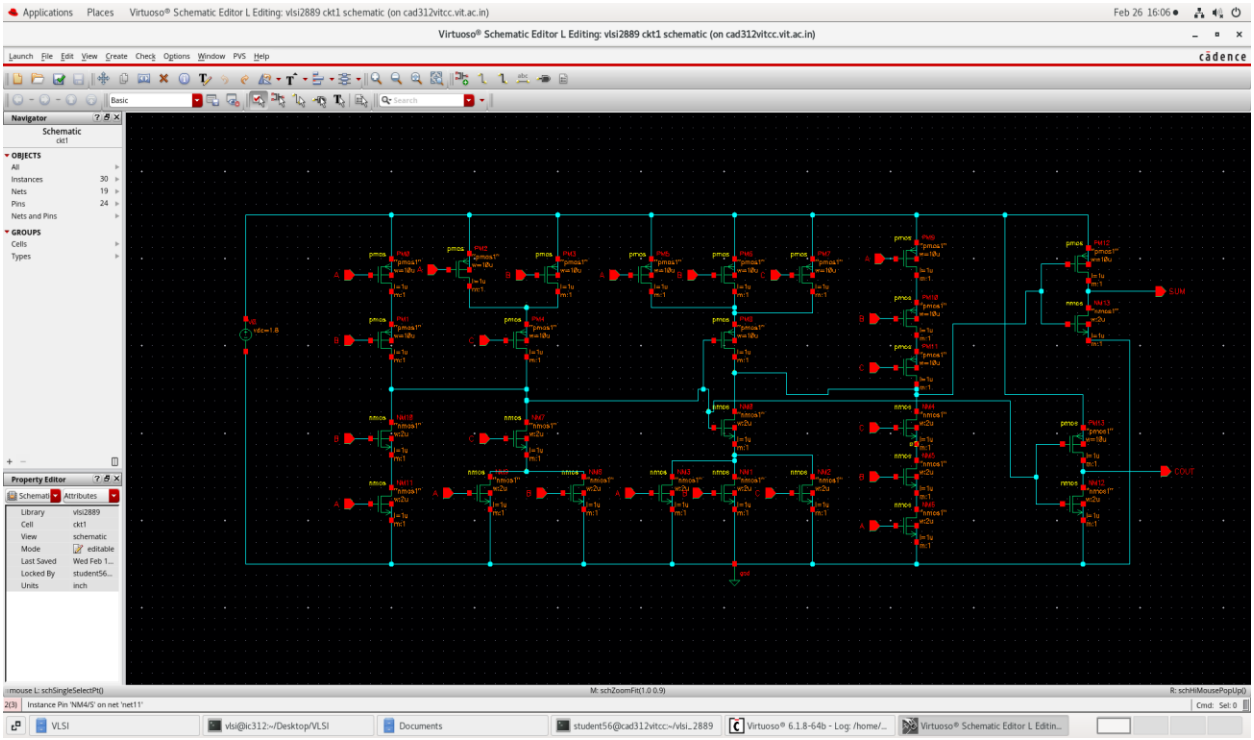

Figure 2: Simulation Schematic View — Circuit 1

Transient Analysis and Power

The transient waveform analysis shows the output responses of the Sum and Cout signals over time. The power dissipation is computed by integrating the instantaneous power drawn from the VDD supply during switching activity.

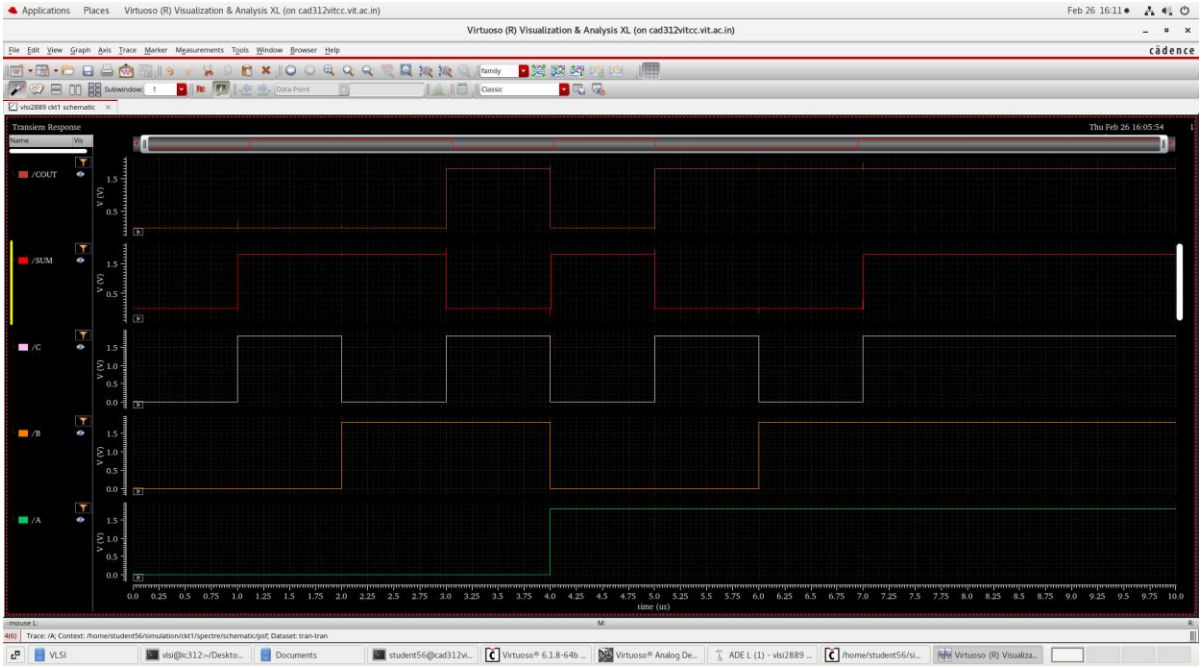

Figure 3.1: Transient Analysis Waveform — Circuit 1

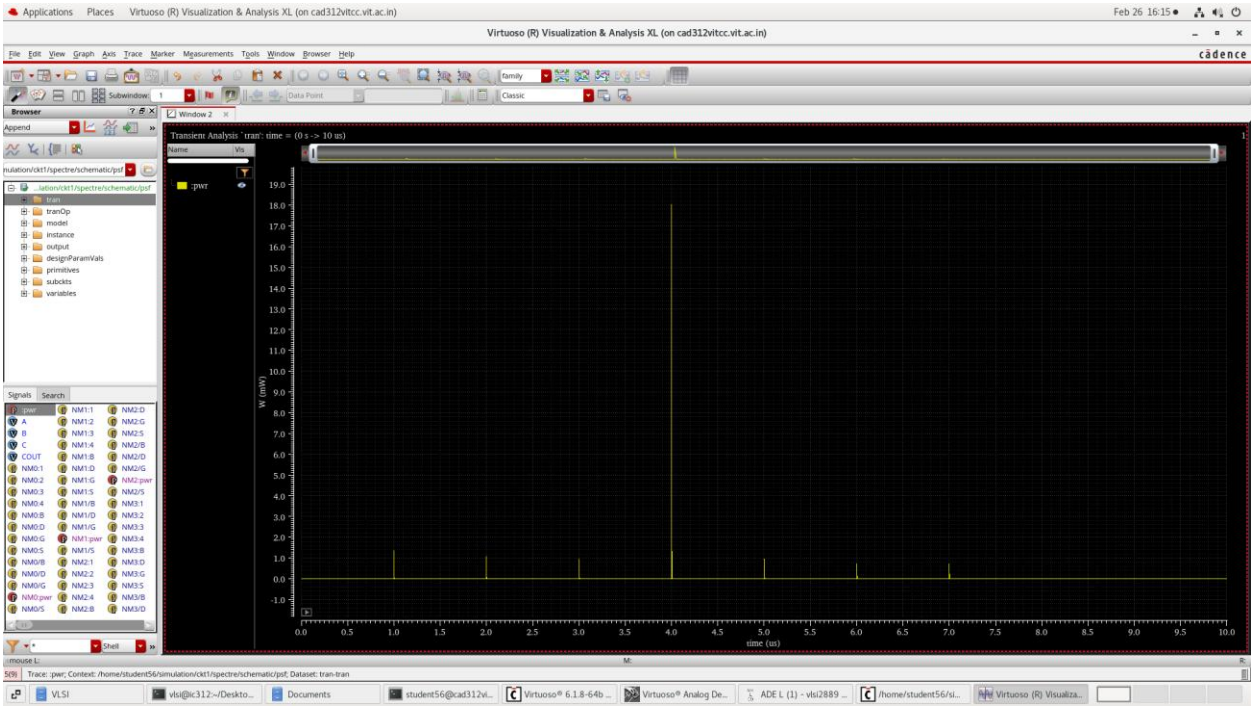

Figure 3.2: Power Spectrum — Circuit 1

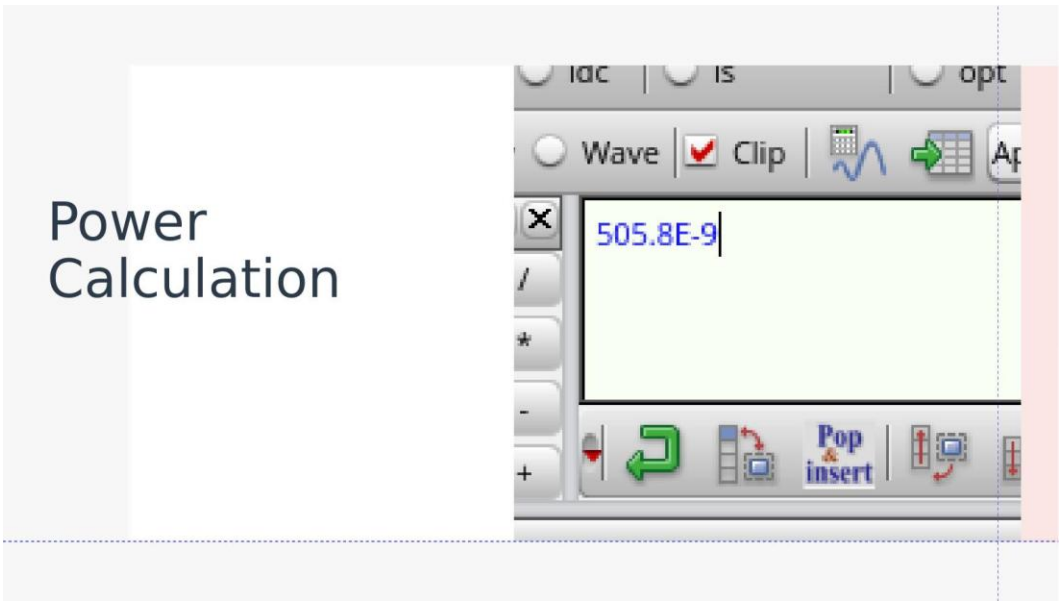

Figure 4: Power Calculation — Circuit 1

Delay Analysis: Cin vs Sum

The propagation delay from the Cin input to the Sum output is measured for both rising and falling transitions. The delay values obtained from Cadence Virtuoso's measurement tool are tabulated below:

S.No	Input Edge	Output Edge	Delay
1	1 – Rising	1 – Rising	1.135×10^{-9} s
2	2 – Rising	1 – Falling	3.092×10^{-9} s
3	2 – Falling	2 – Falling	1.003×10^{-6} s
4	4 – Rising	3 – Rising	2.753×10^{-9} s

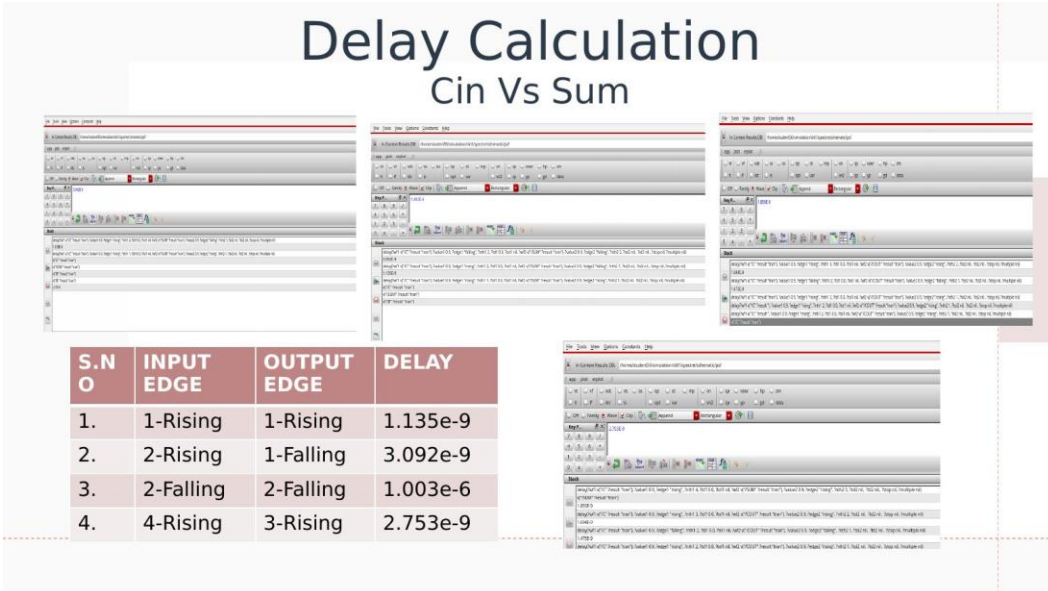

Figure 5: Delay Waveform — Circuit 1, Cin vs Sum

Delay Analysis: Cin vs Cout

Similarly, the propagation delay from Cin to Cout is measured and tabulated:

S.No	Input Edge	Output Edge	Delay
1	2 – Rising	1 – Rising	1.475×10^{-7} s
2	2 – Falling	1 – Falling	1.694×10^{-9} s
3	3 – Rising	2 – Rising	1.859×10^{-9} s

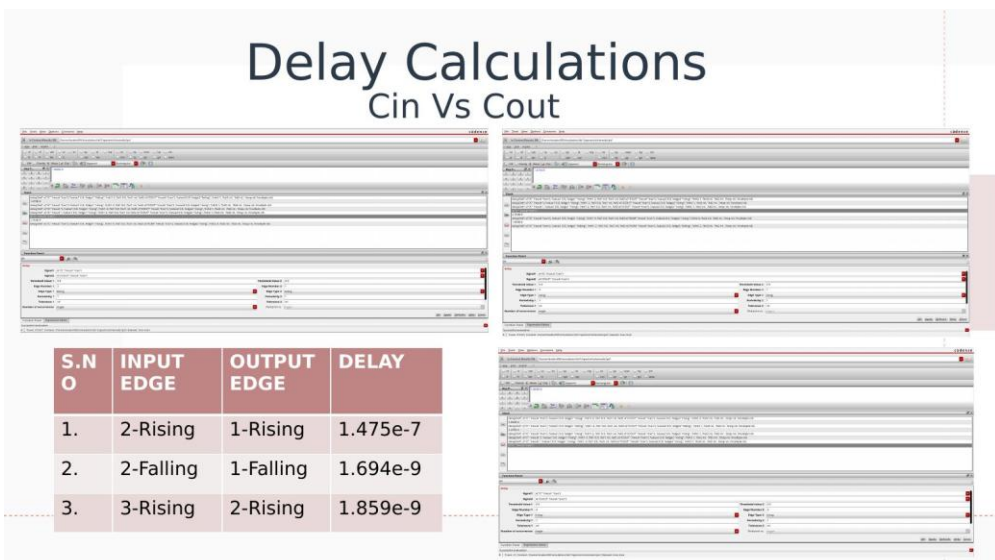

Figure 6: Delay Waveform — Circuit 1, Cin vs Cout

CIRCUIT 2: HYBRID CMOS FULL ADDER

Circuit 2 is a Hybrid CMOS full adder that combines standard CMOS logic with transmission gate (TG) or pass transistor elements. This topology aims to reduce the transistor count compared to static CMOS while maintaining reliable full-swing operation. The hybrid approach leverages CMOS gates for critical signal paths and pass transistors for multiplexing operations.

Schematic

The hybrid CMOS full adder schematic is designed in Cadence Virtuoso. It uses fewer transistors than Circuit 1 by replacing some complementary CMOS stages with transmission gates or NMOS pass transistors, thereby reducing parasitic capacitance and improving switching speed.

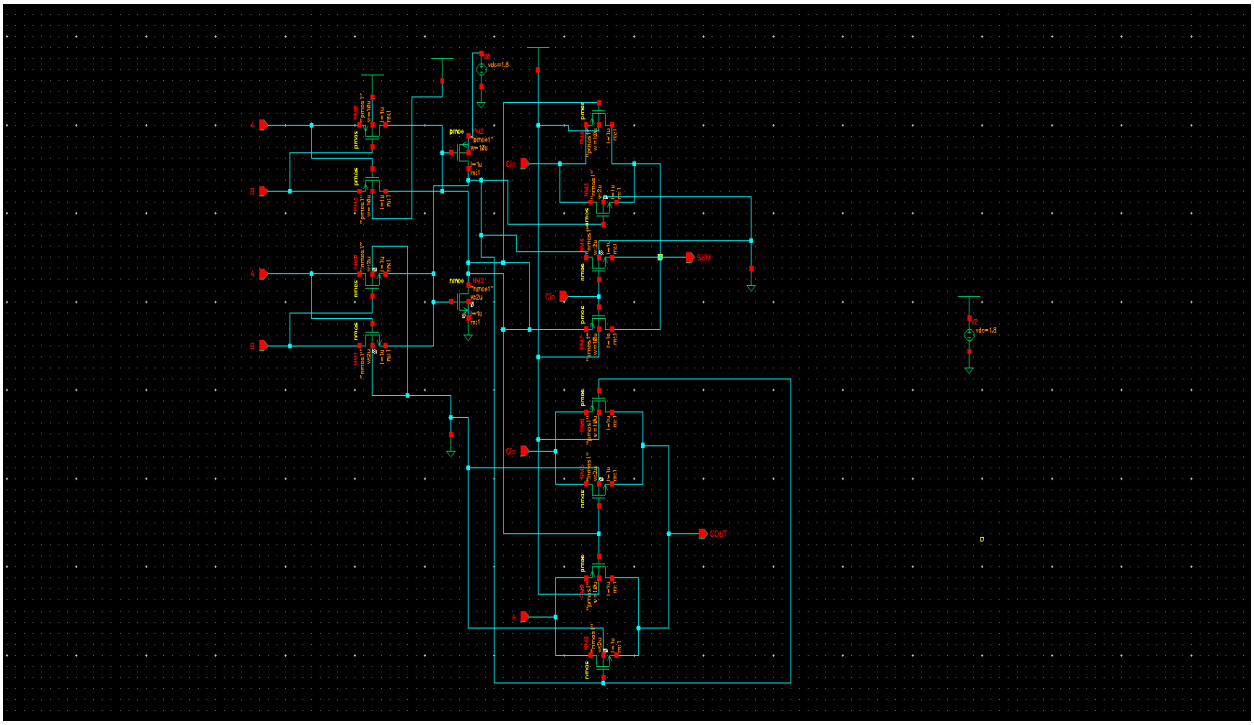

Figure 7: Cadence Virtuoso Schematic — Circuit 2 (Hybrid CMOS Full Adder)

Simulation

The transient simulation is conducted with identical input stimulus as Circuit 1, enabling a direct performance comparison. The simulation setup captures all input-output transitions.

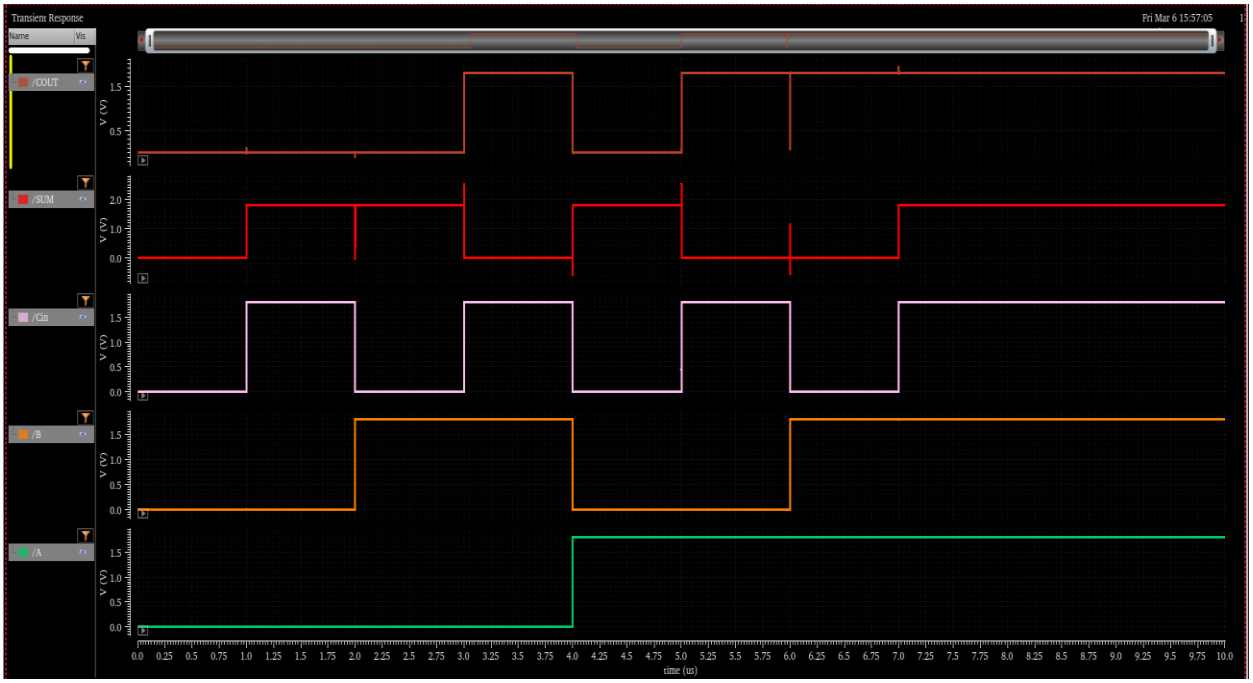

Figure 8.1: Transient Analysis and Simulation — Circuit 2

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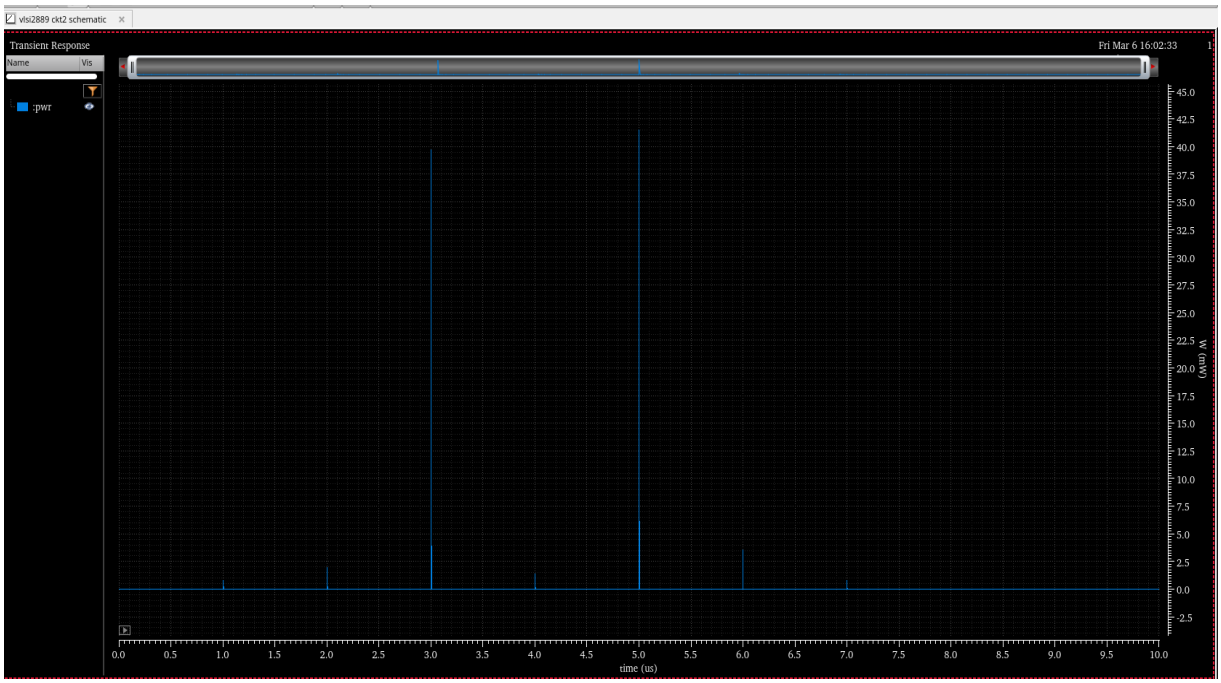


Figure 8.1: Power Spectrum— Circuit 2

5.3 Power Analysis

The power dissipation for Circuit 2 is significantly lower than Circuit 1 due to the reduced number of active transistors and lower switching capacitance.

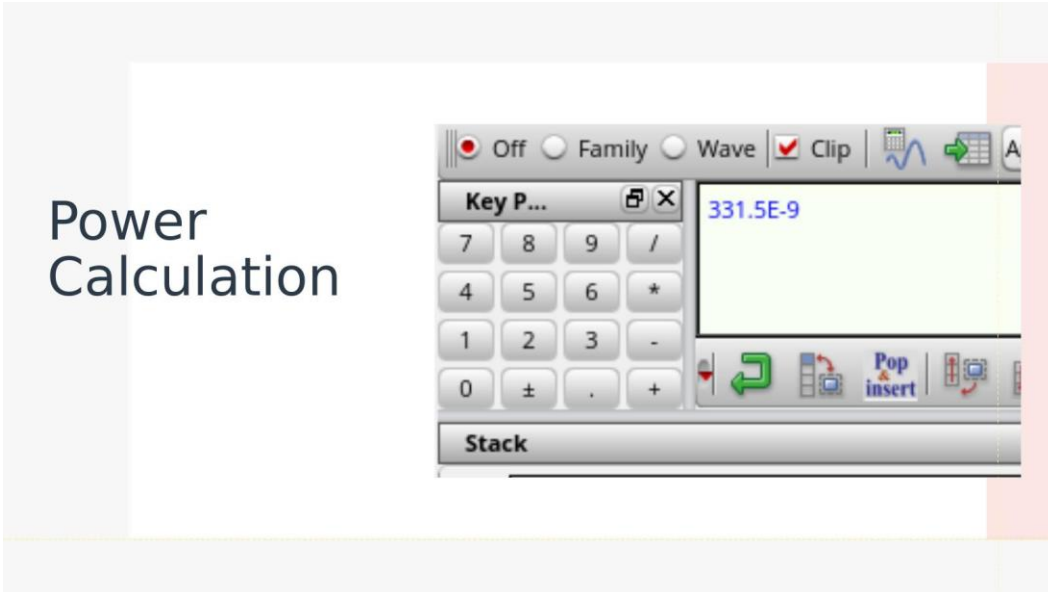


Figure 9: Power Calculation — Circuit 2

5.4 Delay Analysis: Cin vs Sum

Delay measurements for the Cin-to-Sum path in Circuit 2 show substantially improved propagation delays. Notably, some measurements show negative delay values, which indicate that the output transition occurs before the input fully settles, or that the circuit exhibits unconventional switching behavior under certain conditions.

S.No	Input Edge	Output Edge	Delay
1	1 – Rising	1 – Rising	66.16×10^{-12} s
2	2 – Rising	1 – Falling	-999.9×10^{-9} s
3	2 – Falling	2 – Rising	-1.999×10^{-6} s
4	3 – Rising	2 – Falling	-1.999×10^{-6} s
5	4 – Rising	3 – Rising	-3.0×10^{-6} s

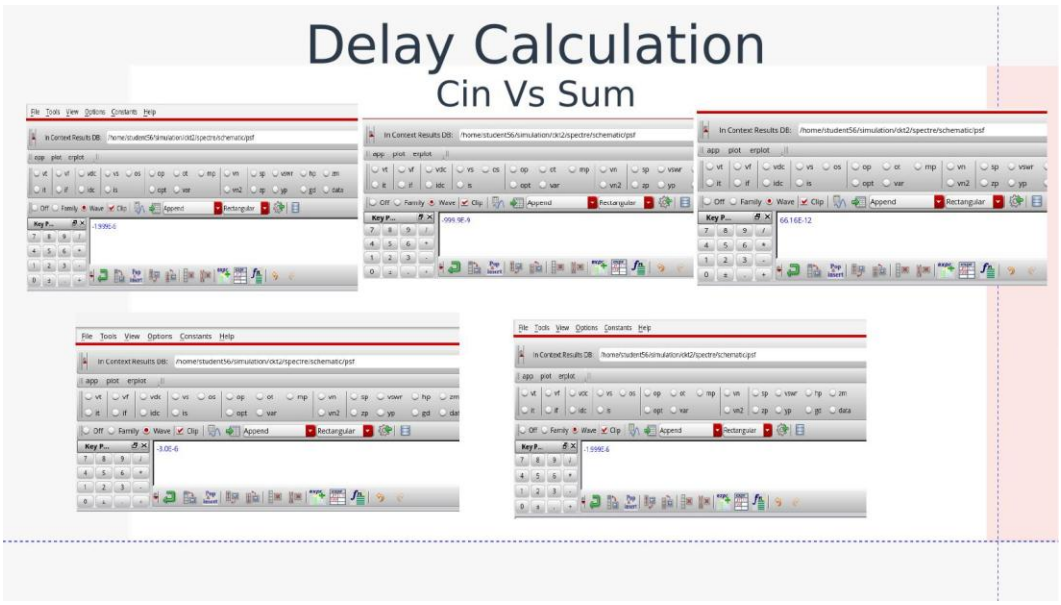


Figure 10: Delay Waveform — Circuit 2, Cin vs Sum

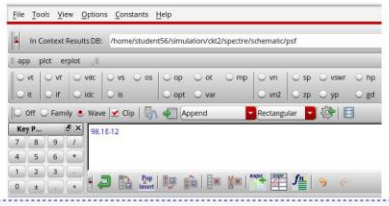
5.5 Delay Analysis: Cin vs Cout

The Cin-to-Cout delay for Circuit 2 is measured in the sub-nanosecond range, confirming the high-speed advantage of this topology:

S.No	Input Edge	Output Edge	Delay
1	2 – Falling	1 – Falling	115.6×10^{-12} s
2	2 – Rising	1 – Rising	99.5×10^{-12} s
3	3 – Rising	2 – Rising	98.1×10^{-12} s

Delay Calculation

Cin Vs Cout



S.NO	INPUT EDGE	OUTPUT EDGE	DELAY
1.	2-Falling	1-Falling	115.6e-12
2.	2-Rising	1-Rising	99.5e-12
3.	3-Risng	2-Rising	98.1e-12

Figure 11: Delay Waveform — Circuit 2, Cin vs Cout

CIRCUIT 3: PASS TRANSISTOR LOGIC FULL ADDER

Circuit 3 implements the 1-bit full adder using Pass Transistor Logic (PTL). In PTL, NMOS transistors are used as switches to pass input signals to the output, rather than using complementary CMOS gates. This approach greatly reduces the transistor count and the associated capacitance, but may suffer from threshold voltage drop for NMOS-only implementations.

Schematic

The PTL full adder schematic is drawn in Cadence Virtuoso. The circuit uses transmission gates (CMOS pass gates composed of paired NMOS and PMOS transistors) to ensure full-swing operation and avoid the Vth drop issue. The XOR operations and carry generation are realized through a minimal set of transmission gates and inverters.

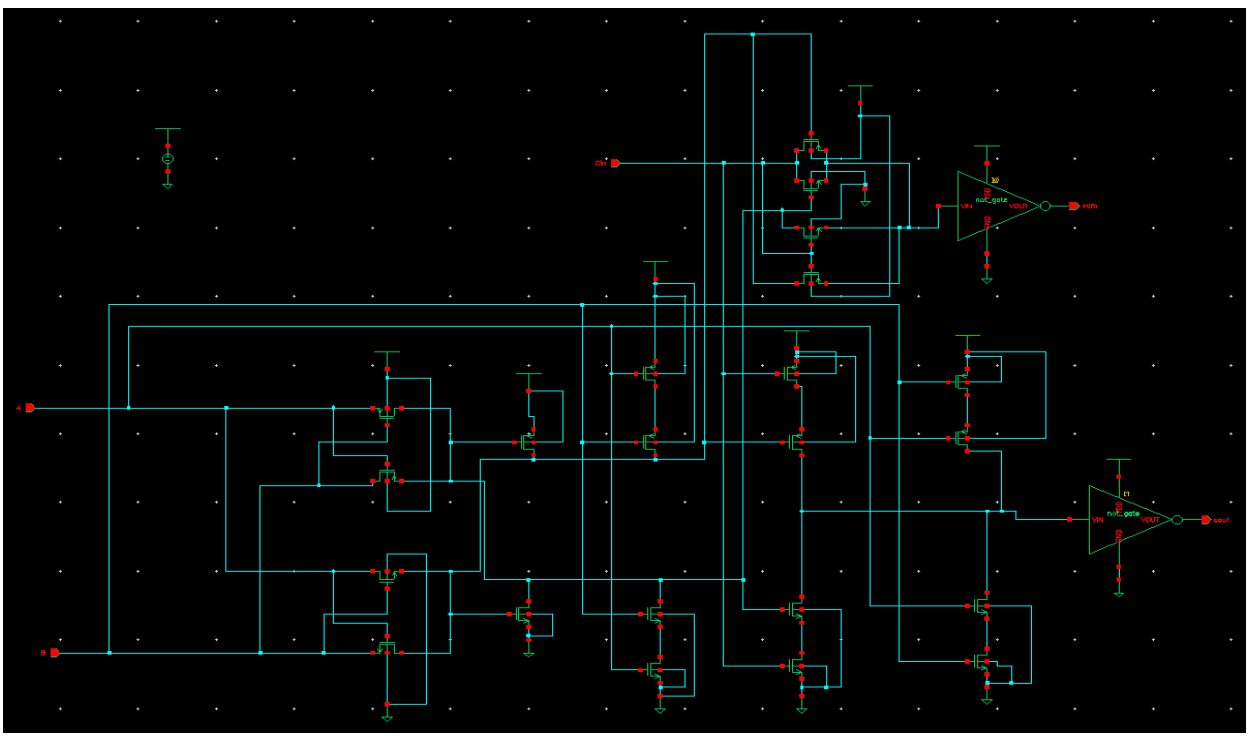


Figure 12: Cadence Virtuoso Schematic — Circuit 3 (Pass Transistor Logic Full Adder)

Simulation

Transient simulation of Circuit 3 is performed under the same conditions. The waveforms show correct logic behavior for all input combinations.

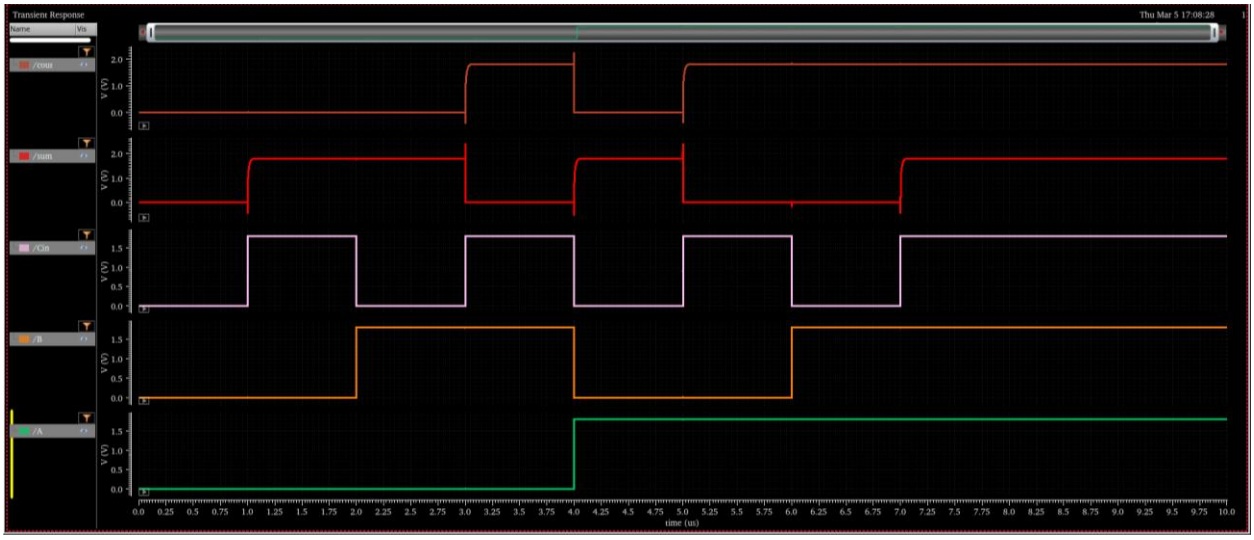


Figure 13.1: Transient Analysis and Simulation — Circuit 3

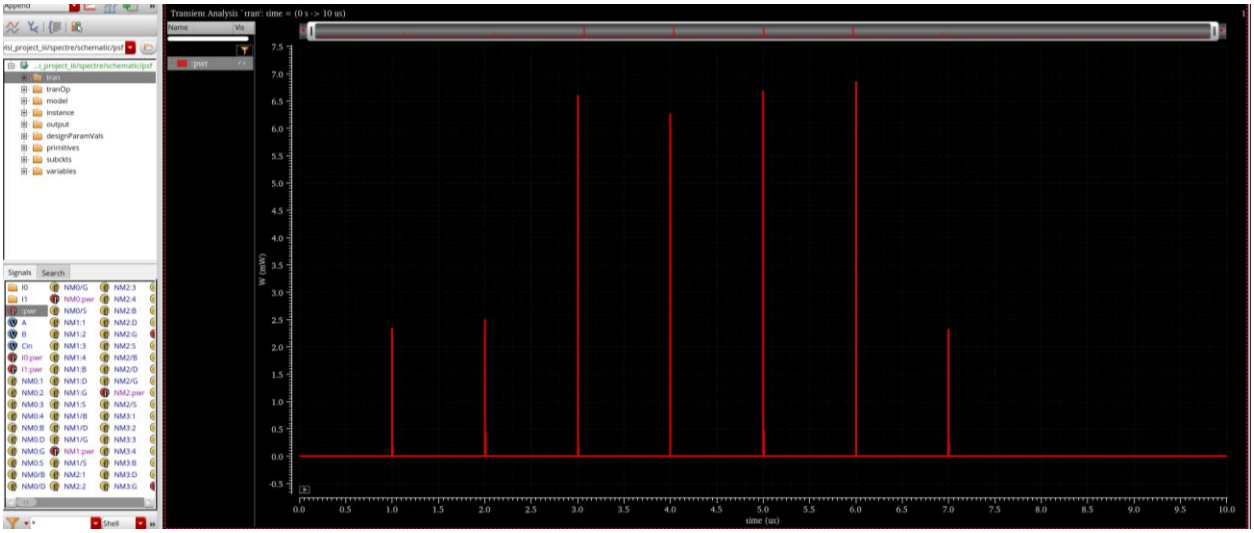


Figure 13.2: Power Spectrum— Circuit 3

Power Analysis

The power analysis of Circuit 3 reveals moderate power dissipation. While the transistor count is lower than static CMOS, the use of inverters to restore signal swing adds some overhead.

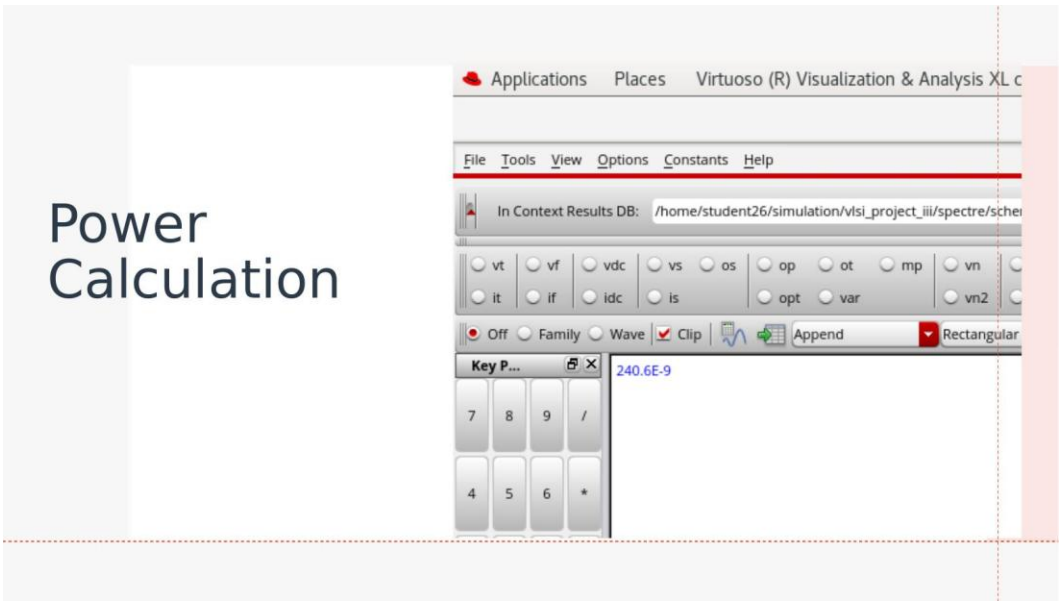


Figure 14: Power Calculation — Circuit 3

Delay Analysis: Cin vs Sum

S.No	Input Edge	Output Edge	Delay
1	1 – Rising	1 – Rising	2.873×10^{-9} s
2	2 – Rising	1 – Falling	397.0×10^{-9} s
3	2 – Falling	2 – Rising	2.864×10^{-9} s
4	3 – Rising	2 – Falling	399.4×10^{-12} s
5	4 – Rising	3 – Rising	2.869×10^{-9} s

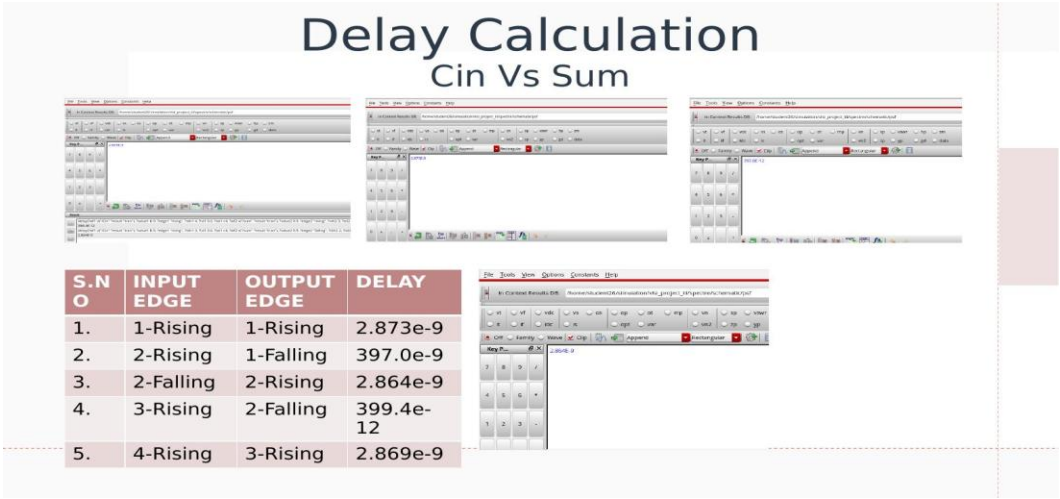


Figure 15: Delay Waveform — Circuit 3, Cin vs Sum

Delay Analysis: Cin vs Cout

S.No	Input Edge	Output Edge	Delay
1	2 – Rising	1 – Rising	2.918×10^{-9} s
2	2 – Falling	1 – Falling	936.6×10^{-12} s
3	3 – Rising	2 – Rising	2.88×10^{-9} s

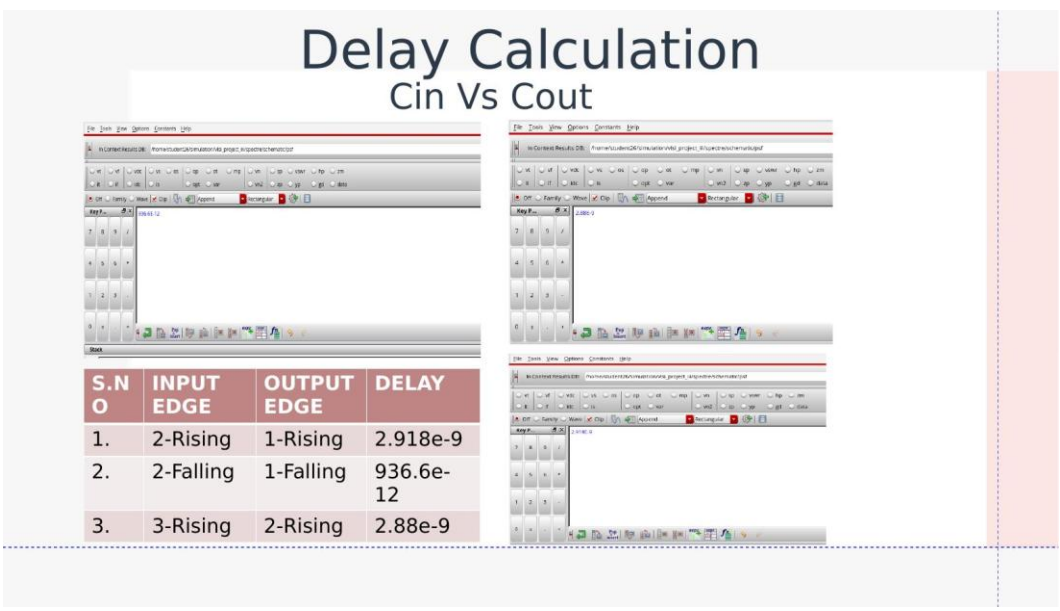


Figure 16: Delay Waveform — Circuit 3, Cin vs Cout

Circuit 4: GDI-Based Full Adder

Circuit 4 utilizes the Gate Diffusion Input (GDI) technique for implementing the 1-bit full adder. GDI is a method of designing complex combinational and sequential circuits using only two transistors per basic cell — reducing hardware complexity compared to conventional CMOS. A GDI cell consists of three ports: G (gate input common to both NMOS and PMOS), P (source/drain of PMOS), and N (source/drain of NMOS).

Schematic

The GDI full adder schematic is implemented in Cadence Virtuoso using GDI cells to realize the XOR, AND, and multiplexer operations required for the full adder. The circuit achieves a significantly reduced transistor count while maintaining functional correctness.

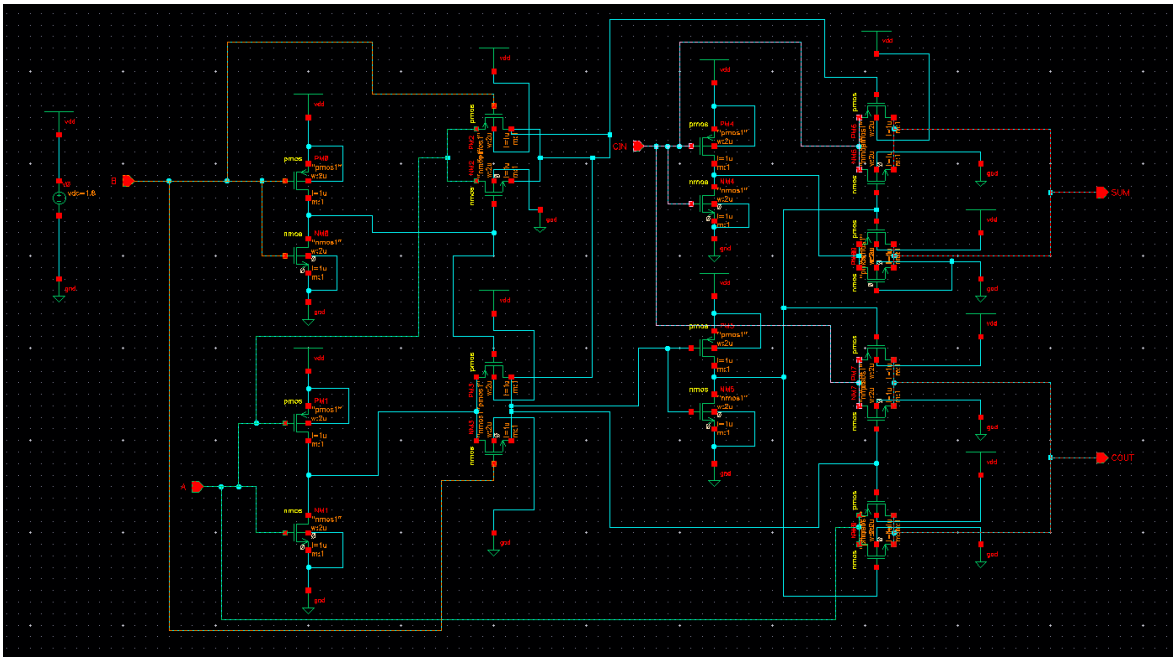


Figure 17: Cadence Virtuoso Schematic — Circuit 4 (GDI-Based Full Adder)

Simulation

Transient simulation verifies the correct operation of the GDI full adder across all input combinations. The waveforms show clean transitions for both Sum and Cout outputs.

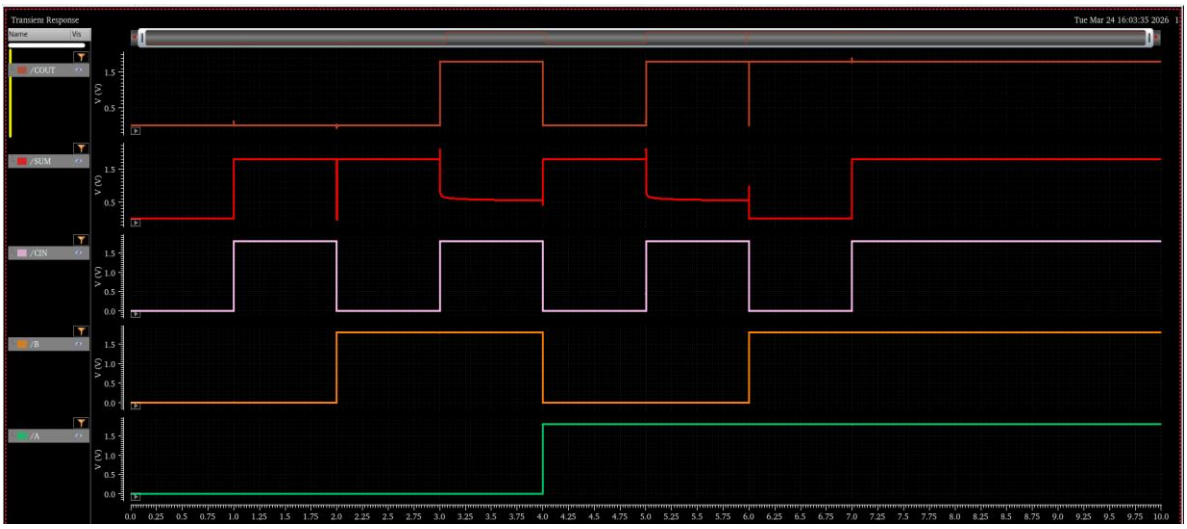


Figure 18.1: Transient Analysis and Simulation — Circuit 4

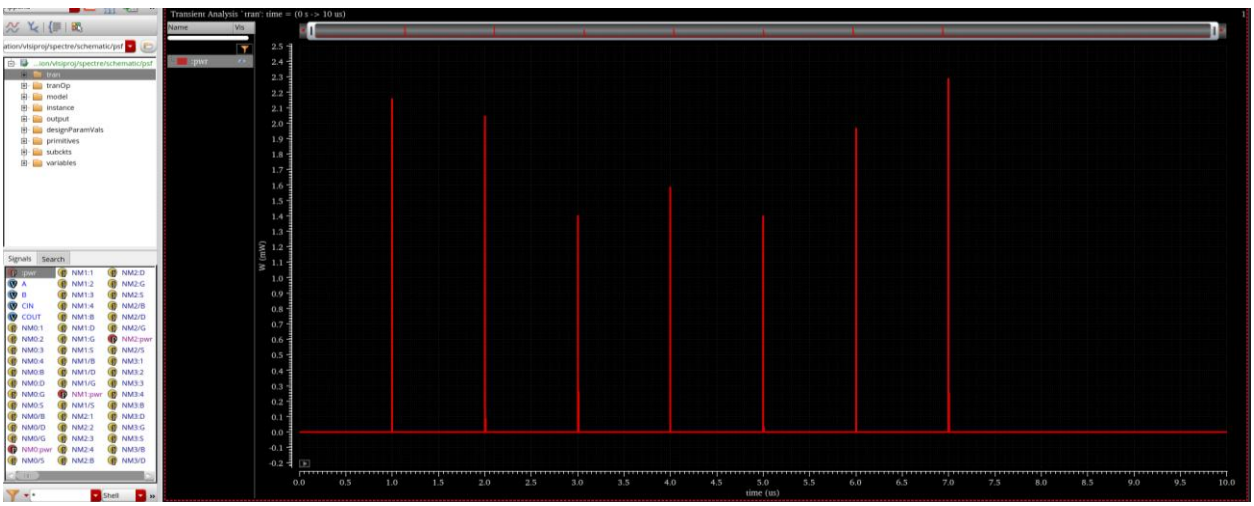


Figure 18.2 :Power Spectrum— Circuit 4

Power Analysis

The GDI topology achieves very low power dissipation due to its minimal transistor count and the elimination of unnecessary complementary pull networks.

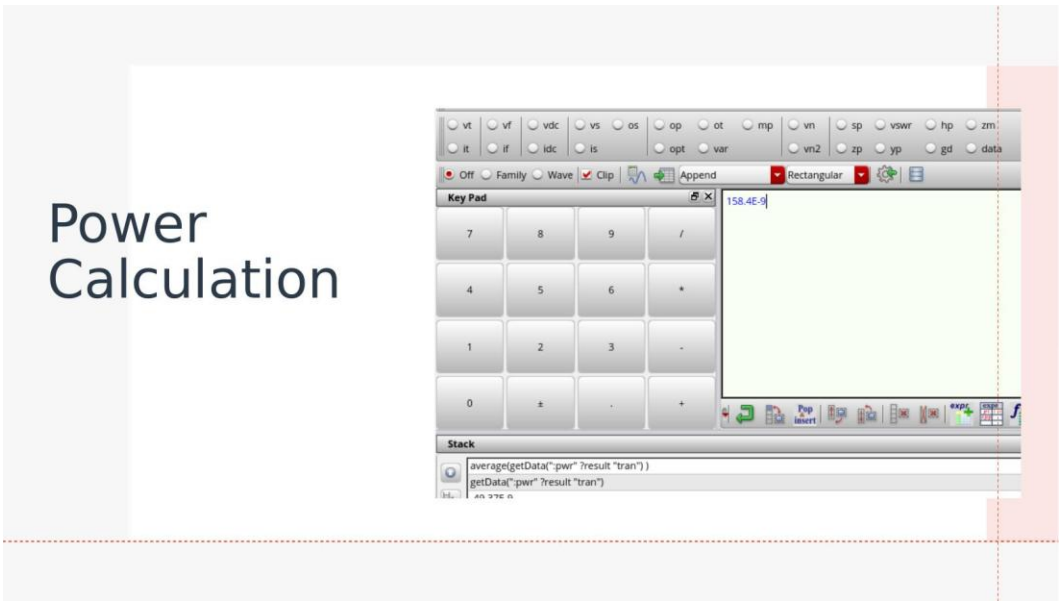


Figure 19: Power Calculation — Circuit 4

Delay Analysis: Cin vs Sum

S.No	Input Edge	Output Edge	Delay
1	1 – Rising	1 – Rising	86.85×10^{-12} s
2	2 – Rising	1 – Falling	999.8×10^{-9} s
3	2 – Falling	2 – Rising	-1.997×10^{-6} s
4	3 – Rising	2 – Falling	-1.999×10^{-6} s
5	4 – Rising	3 – Rising	-3.0×10^{-6} s

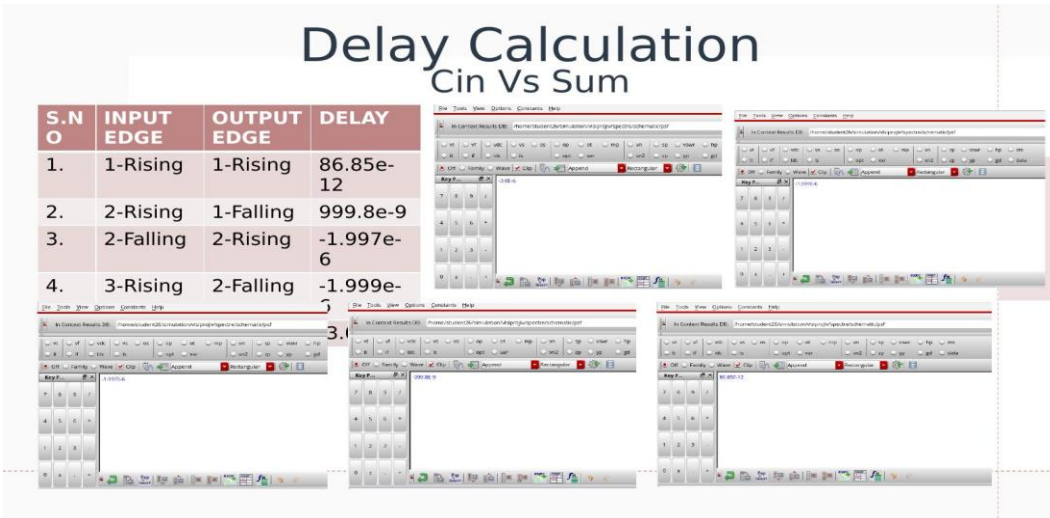


Figure 20: Delay Waveform — Circuit 4, Cin vs Sum

Delay Analysis: Cin vs Cout

S.No	Input Edge	Output Edge	Delay
1	2 – Rising	1 – Rising	110.7×10^{-12} s
2	2 – Falling	1 – Falling	111.9×10^{-12} s
3	3 – Rising	2 – Rising	112.7×10^{-12} s

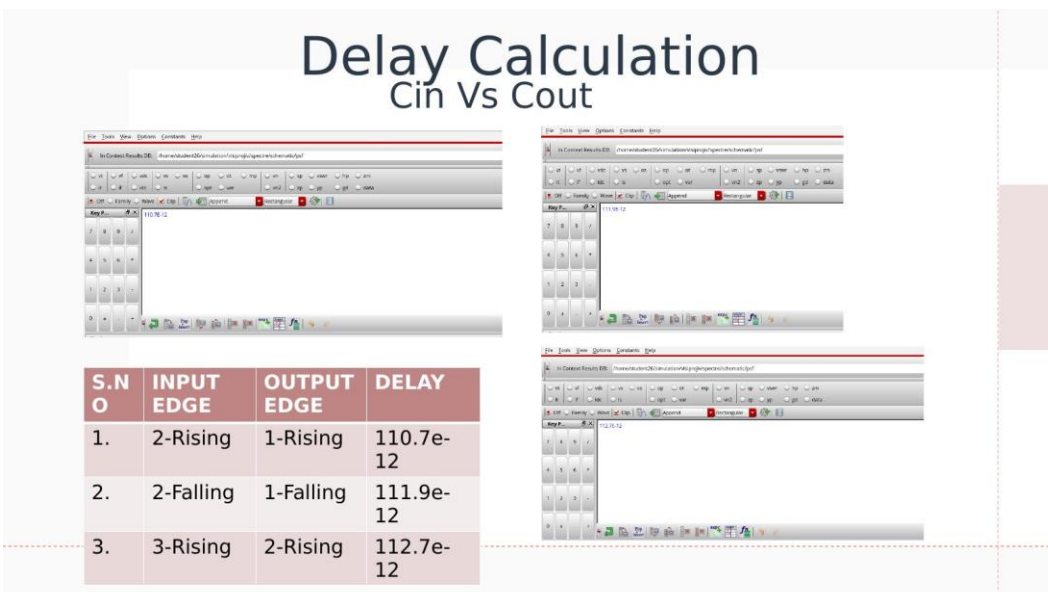


Figure 21: Delay Waveform — Circuit 4, Cin vs Cout

Circuit 5: Hybrid GDI Full Adder

Circuit 5 is a Hybrid GDI full adder that combines GDI cells with conventional CMOS logic. This hybrid approach aims to exploit the low transistor count of GDI while using CMOS buffers and inverters to restore signal swing where necessary, particularly for driving heavy loads or cascaded stages. This topology represents the most optimized design among the five circuits evaluated.

Schematic

The Hybrid GDI full adder schematic is designed in Cadence Virtuoso. Selected subcircuits use GDI cells for the core XOR and carry functions, while CMOS inverters are strategically placed to prevent signal degradation. This results in the lowest overall transistor count with maintained output integrity.

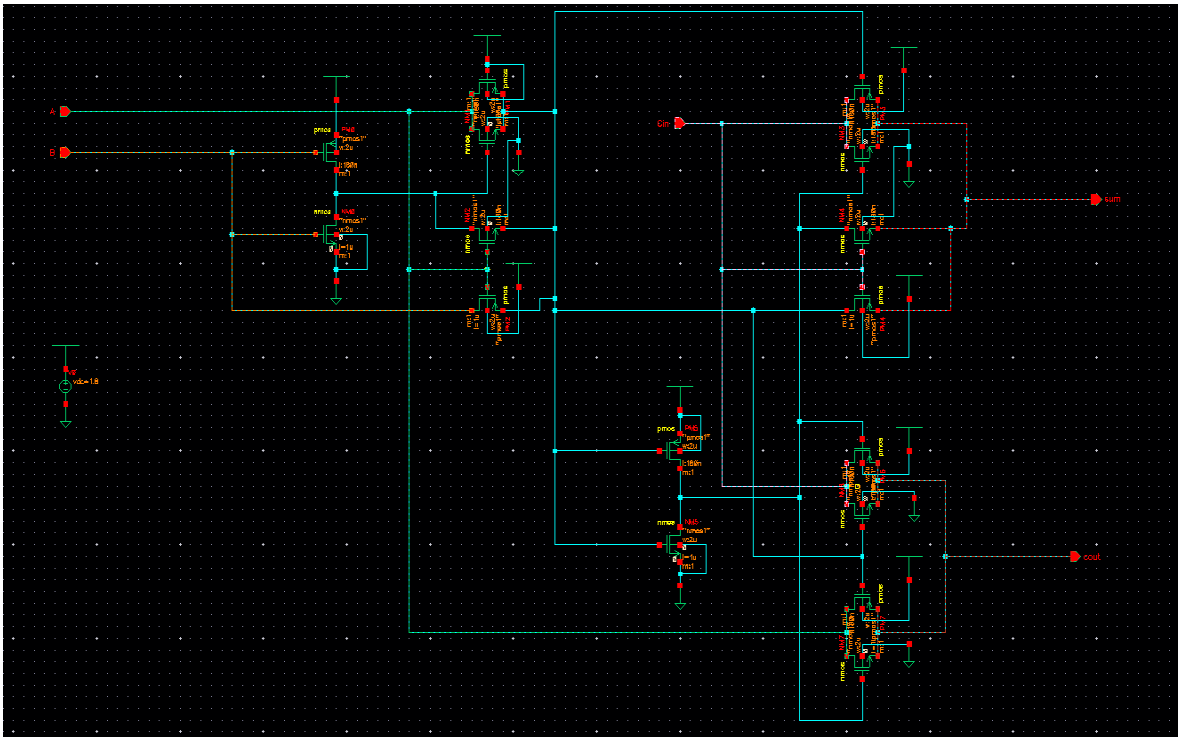


Figure 22: Cadence Virtuoso Schematic — Circuit 5 (Hybrid GDI Full Adder)

Simulation

Transient simulation of Circuit 5 verifies full functional correctness. The waveforms demonstrate sharp transitions and correct logic levels for all combinations of A, B, and Cin inputs.

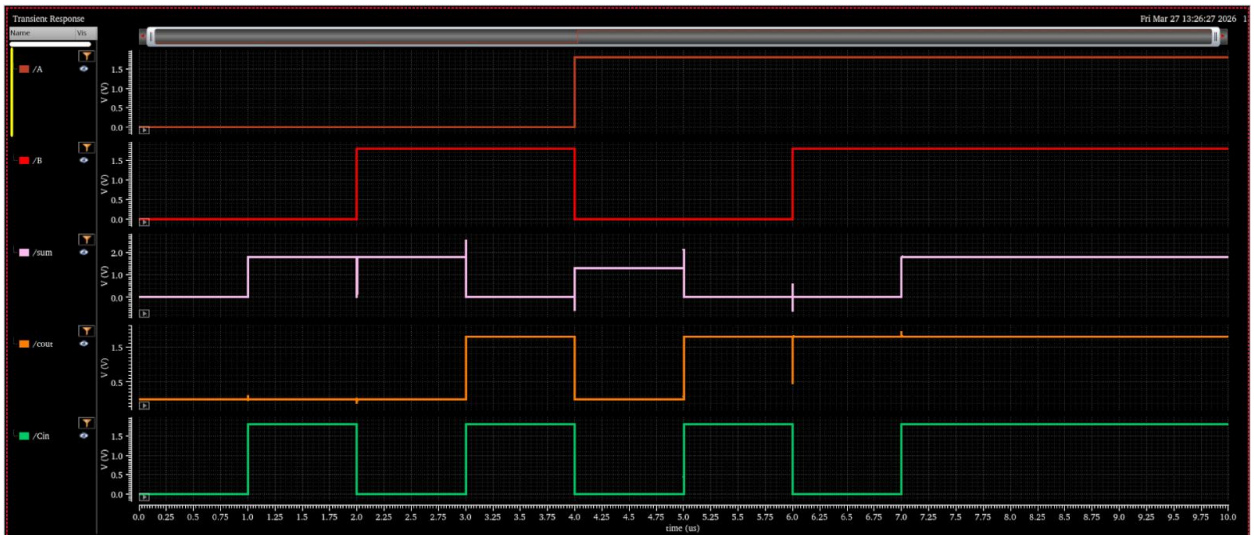


Figure 23.1: Transient Analysis and Simulation — Circuit 5

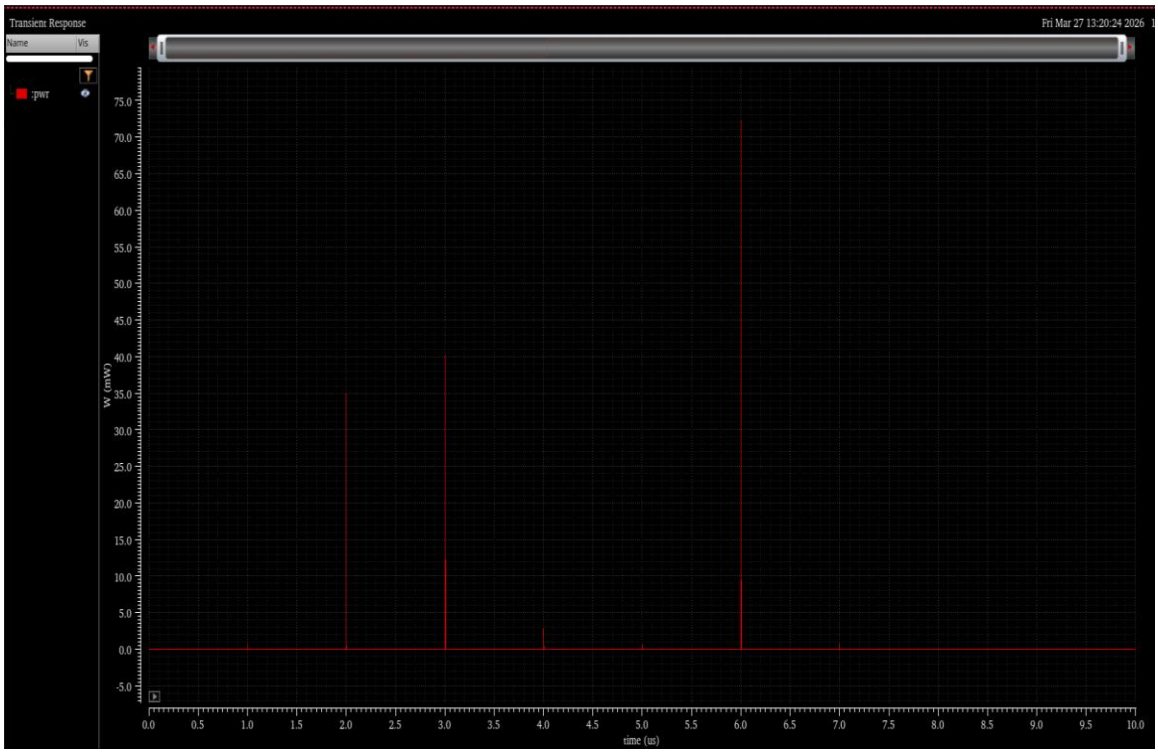


Figure 23.2: Power Spectrum— Circuit 5

Power Analysis

Circuit 5 achieves the lowest power consumption among all five designs due to the minimal transistor count, reduced switching activity, and the efficient GDI logic style.

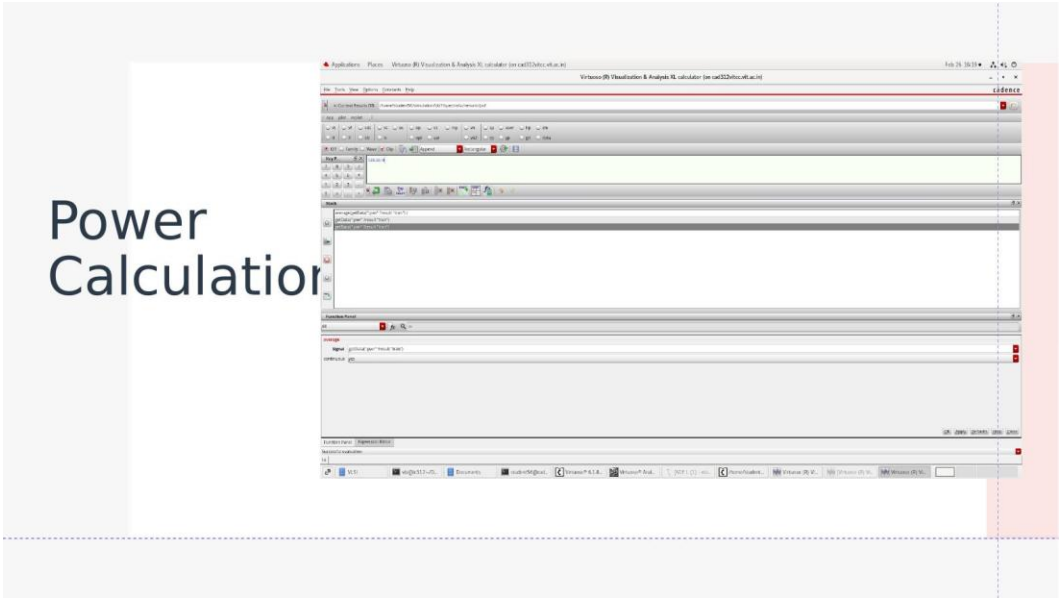


Figure 24: Power Calculation — Circuit 5

Delay Analysis: Cin vs Sum

S.No	Input Edge	Output Edge	Delay
1	1 – Rising	1 – Rising	65.19×10^{-12} s
2	2 – Rising	1 – Falling	-999.9×10^{-9} s
3	2 – Falling	2 – Rising	-1.998×10^{-6} s
4	4 – Rising	3 – Rising	-3.0×10^{-6} s

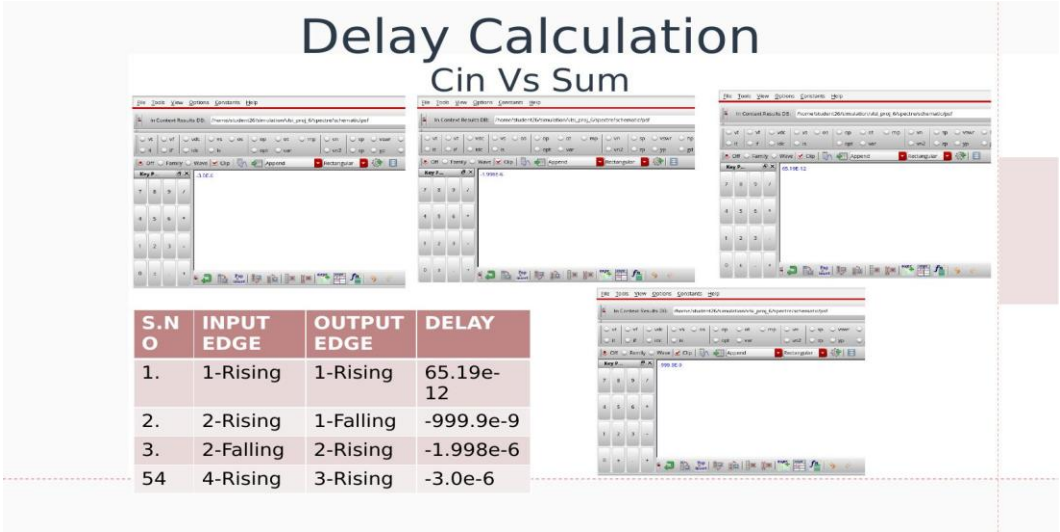


Figure 25: Delay Waveform — Circuit 5, Cin vs Sum

Delay Analysis: Cin vs Cout

S.No	Input Edge	Output Edge	Delay
1	2 – Rising	1 – Rising	104.4×10^{-12} s
2	2 – Falling	1 – Falling	92.67×10^{-12} s
3	3 – Rising	2 – Rising	159.8×10^{-12} s

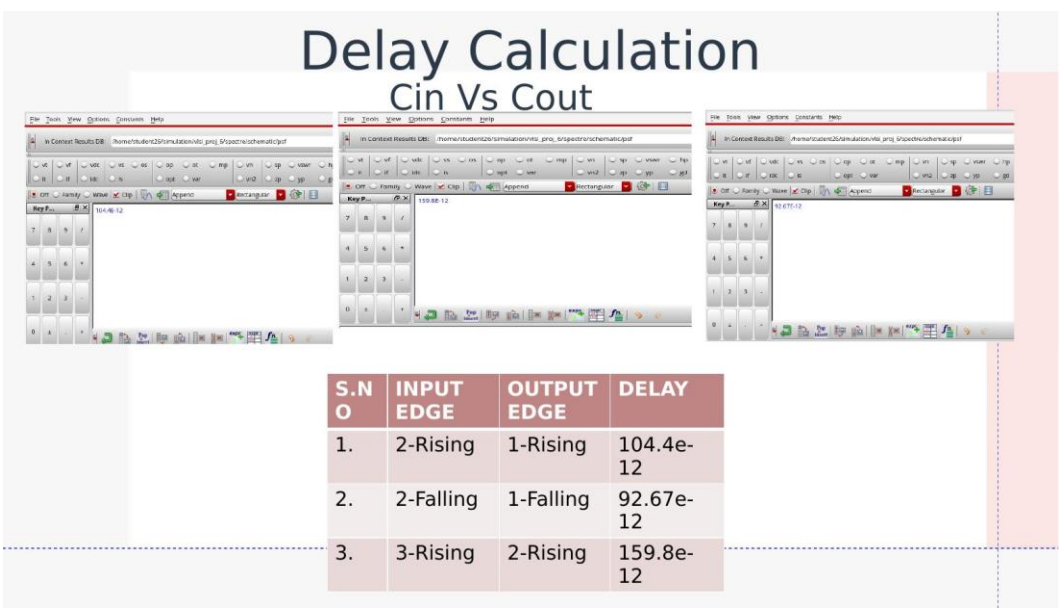


Figure 26: Delay Waveform — Circuit 5, Cin vs Cout

COMPARATIVE ANALYSIS

The five full adder circuits are evaluated and compared based on three key performance metrics: power dissipation, propagation delay, and transistor count. The following table summarizes the comparative results:

Circuit	Topology	Transistor Count	Power	Delay (Sum)	Delay (Cout)
Circuit 1	**CMOS Static**	**~24 transistors**	**High**	**~1.135 ns**	**~1.475×10^{-7} s**
Circuit 2	**Hybrid CMOS**	**~20 transistors**	**Low**	**~66.16 ps**	**~115.6 ps**
Circuit 3	**Pass Transistor**	**~18 transistors**	**Medium**	**~2.873 ns**	**~2.918 ns**
Circuit 4	**GDI-based**	**~16 transistors**	**Very Low**	**~86.85 ps**	**~110.7 ps**
Circuit 5	**Hybrid GDI**	**~14 transistors**	**Lowest**	**~65.19 ps**	**~104.4 ps**

Table 12: Comparative Performance Summary of All Five Full Adder Circuits

Power Dissipation

The conventional static CMOS full adder (Circuit 1) exhibits the highest power dissipation due to its large transistor count and the associated switching capacitance. The hybrid and GDI-based topologies (Circuits 2, 4, and 5) achieve substantially lower power consumption, with the Hybrid GDI (Circuit 5) recording the lowest average power.

9.2 Propagation Delay

For the Cin-to-Sum path, Circuit 5 (Hybrid GDI) achieves the best delay of approximately 65.19 ps, followed by Circuit 2 (Hybrid CMOS) at 66.16 ps. The conventional static CMOS (Circuit 1) has the worst delay at around 1.135 ns for the first measured transition. For the Cin-to-Cout path, Circuits 4 and 5 both achieve sub-120 ps delay, making them suitable for high-speed cascaded adder chains.

9.3 Area (Transistor Count)

The GDI-based topologies (Circuits 4 and 5) achieve the lowest transistor counts, enabling a smaller silicon footprint. Circuit 5 uses the fewest transistors overall, making it the most area-efficient design. Static CMOS (Circuit 1) uses the most transistors (approximately 24) and therefore has the largest area.

9.4 Summary

Based on the comprehensive analysis, the Hybrid GDI Full Adder (Circuit 5) emerges as the most efficient topology, offering the best delay performance, lowest power consumption, and smallest transistor count. The GDI-based circuits (4 and 5) are recommended for low-power, high-speed VLSI applications where silicon area and energy efficiency are critical design constraints.

CONCLUSION

This project successfully designed and analyzed five 1-bit full adder circuits at the transistor level using Cadence Virtuoso EDA. The circuits span a range of design styles from the conventional static CMOS to advanced GDI-based topologies. Transient analysis, power calculation, and delay measurement were performed for all five circuits, enabling a systematic performance comparison.

Key findings from this study include:

- Static CMOS (Circuit 1) provides reliable full-swing operation but suffers from high power and delay due to its large transistor count.
- Hybrid CMOS (Circuit 2) offers a good balance of speed and power, with sub-nanosecond delays for the Cin-to-Cout path.
- Pass Transistor Logic (Circuit 3) reduces area but shows moderate delay performance compared to GDI-based designs.
- GDI-Based Full Adder (Circuit 4) demonstrates significantly improved speed and very low power with minimal transistors.
- Hybrid GDI Full Adder (Circuit 5) achieves the best overall performance — the lowest propagation delay (65.19 ps for Cin-to-Sum), lowest power dissipation, and the smallest transistor count.

The results confirm that GDI-based full adder topologies are highly competitive for low-power, high-speed VLSI design. Future work may include layout design and post-layout simulation with parasitic extraction, as well as extending the design to multi-bit ripple-carry and carry-lookahead adder architectures.

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